

- (iii) If $\nu = \mathcal{T}$ (the article is truthful), then its message is generated as $m = \theta$ with probability $p > 1/2$. Conversely, if $\nu = \mathcal{M}$ (the article contains misinformation), then its message is generated as $m = \theta$ with probability $q \leq 1/2$ and is weakly anti-correlated with the truth.

While m and r are common knowledge (for example, the message m is directly observed and reliability depends on certain commonly-observed characteristics such as source and headline), the third dimension, ν , is unknown to all agents. We assume that agents update their beliefs about ν using Bayes' rule given beliefs about the underlying state θ and the observables (r, m) of the article.

Social Media Behavior. Time is discrete $t = 1, 2, \dots$ Upon receipt of the article, an agent i can take one of three actions $a_i \in \{\mathcal{S}, \mathcal{I}, \mathcal{D}\}$, as described below:

- (i) Share ($\mathcal{S}$): The agent decides to *share* the article and passes it onto others after her.
- (ii) Ignore ($\mathcal{I}$): The agent decides to *ignore* the article and does not engage with it.
- (iii) Dislike ($\mathcal{D}$): The agent decides to *dislike* the article, which means expressing disagreement or contempt for the content contained in it.

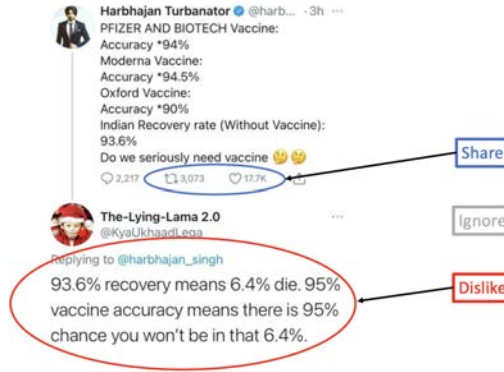


Figure 1. Sample Tweet.

The three possible actions are depicted in Figure 1 using Twitter as a sample social media platform. A given user sees the article and decides how to respond to it. She can (i) share it ($\mathcal{S}$), which actively puts it on other social media news feeds; (ii) ignore it ($\mathcal{I}$), where the user simply scrolls past the article; or (iii) actively dislike it ($\mathcal{D}$), expressing derision for the content.

At time $t = 1$, we assume that some initial seed agent i^* begins with the article. If the article is shared by agent i , it is passed to all $j \in \mathcal{N}_i$. In contrast, following ignore or dislike, the article does not propagate past agent i .

Payoffs. Let us define shares after i as $S_i = |\{j \in \mathcal{N}_i : a_j = \mathcal{S}\}|$ and dislikes after i as $D_i = |\{j \in \mathcal{N}_i : a_j = \mathcal{D}\}|$. Agent i 's utility can then be written as

$$U_i = \begin{cases} 0, & \text{if } a_i = \mathcal{I} \\ \tilde{u}1_{\nu=\mathcal{M}} - \tilde{c}, & \text{if } a_i = \mathcal{D} \\ u1_{\nu=\mathcal{T}} - c1_{\nu=\mathcal{M}} + \kappa S_i - dD_i, & \text{if } a_i = \mathcal{S} \end{cases} \quad (1)$$